

A low-fat, whole-food vegan diet, as well as other strategies that down-regulate IGF-I activity, may slow the human aging process.

A considerable amount of evidence is consistent with the proposition that systemic IGF-I activity acts as pacesetter in the aging process. A reduction in IGF-I activity is the common characteristic of rodents whose maximal lifespan has been increased by a wide range of genetic or dietary measures, including caloric restriction. The lifespans of breeds of dogs and strains of rats tend to be inversely proportional to their mature weight and IGF-I levels. The link between IGF-I and aging appears to be evolutionarily conserved; in worms and flies, lifespan is increased by reduction-of-function mutations in signaling intermediates homologous to those which mediate insulin/IGF-I activity in mammals. The fact that an increase in IGF-I activity plays a key role in the induction of sexual maturity, is consistent with a broader role for-IGF-I in aging regulation. If down-regulation of IGF-I activity could indeed slow aging in humans, a range of practical measures for achieving this may be at hand. These include a low-fat, whole-food, vegan diet, exercise training, soluble fiber, insulin sensitizers, appetite suppressants, and agents such as flax lignans, oral estrogen, or tamoxifen that decrease hepatic synthesis of IGF-I. Many of these measures would also be expected to decrease risk for common age-related diseases. Regimens combining several of these approaches might have a sufficient impact on IGF-I activity to achieve a useful retardation of the aging process. However, in light of the fact that IGF-I promotes endothelial production of nitric oxide and may be of especial importance to cerebrovascular health, additional measures for stroke prevention-most notably salt restriction-may be advisable when attempting to down-regulate IGF-I activity as a pro-longevity strategy.